

Near-Square-Root NP-Hardness for Euclidean CVP Would Collapse the Exponential Hierarchy

somewhereafter

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Abstract

Aggarwal, Bennett, Brakerski, Golovnev, Kumar, Li, Peters, Stephens-Davidowitz, and Vaikuntanathan proved a quantitative public-coin protocol placing rank- D Euclidean GapCVP_γ in coAM with complexity $2^{O(D/\gamma^2)}$ up to polynomial factors. We record a consequence at the near-square-root approximation frontier. If, for any fixed $C > 0$, Euclidean

$$\text{GapCVP}_{\sqrt{D}/\log^C D}$$

is NP-hard under polynomial-time promise-preserving many-one reductions, then the exponential hierarchy collapses to its second level. The proof combines the quantitative lattice protocol with the resource-scaled Boppana–Håstad–Zachos theorem of Pavan, Selman, Sengupta, and Vinodchandran.

More generally, for a reduction from inputs of length n producing rank $D(n)$ and approximation $\gamma(D(n))$, the same conclusion follows whenever $D(n)/\gamma(D(n))^2 = \log^{O(1)} n$. Writing $\gamma(D) = \sqrt{D}/f(D)$ exposes the exact protocol frontier: every fixed polylogarithmic loss $f(D) = \log^{O(1)} D$ is conditionally excluded. For rank-regular reductions, the first dimension-only regime not excluded by this argument is the superpolylogarithmic-loss strip $f(D) \notin \log^{O(1)} D$. The genuinely near-square-root portion of that region additionally has $f(D) = D^{o(1)}$. This is a barrier theorem, not an unconditional lower bound and not a construction in the surviving strip.

We also give a constant-sensitive sharpening of the Aharonov–Regev Fourier verifier. For every fixed $\varepsilon > 0$, $\text{GapCVP}_{(2/\pi+\varepsilon)\sqrt{D}}$ lies in $\text{NP} \cap \text{coNP}$. In particular, $\text{GapCVP}_{\sqrt{D}}$ is in $\text{NP} \cap \text{coNP}$, so deterministic Karp NP-hardness at the exact unit square-root factor would imply $\text{NP} = \text{coNP}$. This constant sharpening is proved here; it is not attributed to the literal published statement of Aharonov–Regev.

1 Introduction

The Euclidean closest vector problem asks for the distance from a target $t \in \mathbb{Q}^D$ to a full-rank lattice $L \subseteq \mathbb{R}^D$. Its promise version GapCVP_γ distinguishes

$$\text{dist}(t, L) \leq r \quad \text{from} \quad \text{dist}(t, L) > \gamma(D)r.$$

The square-root scale is a persistent boundary in the complexity of lattice problems, beginning with the classical coAM barrier of Goldreich and Goldwasser [GG00]. A particularly sharp prospective hardness factor is

$$\gamma_C(D) = \frac{\sqrt{D}}{\log^C D}, \tag{1}$$

where $C > 0$ is fixed. This factor is $D^{1/2-o(1)}$ and is asymptotically larger than D^c for every fixed $c < 1/2$. It should be compared with Mira’s formalized NP-hardness theorem at D^c for every fixed $c < 1/2$ [Mir26].

Our main observation is that (1) is already inside a universal quasipolynomial-time **coAM** regime. This fact has a stronger consequence than a construction-specific obstruction: polynomial-time NP-hardness at (1) would collapse the exponential hierarchy. No classification of lattice gadgets, codes, Fourier witnesses, or source compilers is needed.

The ingredients are published. Aggarwal et al. [ABB⁺26] prove the time–approximation tradeoff

$$\text{GapCVP}_\gamma \in \text{coAM-TIME} \left(2^{O(D/\gamma^2)} \text{poly}(M) \right), \quad (2)$$

where M is the target-instance encoding length. Pavan et al. [PSSV07] prove a resource-scaled extension of the Boppana–Håstad–Zachos collapse theorem [BsZ87]: quasipolynomial Arthur–Merlin protocols for coNP collapse the exponential hierarchy to its second level. Substituting (1) into (2) gives complexity $2^{O(\log^{2^C} D)} \text{poly}(M)$, which is quasipolynomial under any polynomial-time reduction.

At the exact unit factor, a sharper barrier is available. The theorem statement of Aharonov and Regev [AR05] suppresses its absolute constant, while their displayed proof explicitly yields the safe factor $100\sqrt{D}$. Section 3 retains the constants in the same verifier and proves membership for every $(2/\pi + \varepsilon)\sqrt{D}$. Since $2/\pi < 1$, exact $\sqrt{D}$ -hardness would imply $\text{NP} = \text{coNP}$. This confirms the endpoint consequence stated in Mira’s draft [Mir26], while making the published-versus-new proof boundary explicit.

The contribution of this note is fourfold.

1. We state and prove the resulting architecture-independent conditional impossibility theorem for near-square-root **GapCVP** hardness.
2. We give the source-sensitive form $D(n)/\gamma(D(n))^2 = \log^{O(1)} n$, avoiding an implicit assumption that target rank and source length are polynomially equivalent.
3. We isolate the first asymptotic strip not covered by the quasipolynomial-protocol argument and separate it from the stronger AMETH barrier already implicit in [ABB⁺26].
4. We prove the constant-sensitive Aharonov–Regev sharpening and its $\text{NP} = \text{coNP}$ consequence at the exact unit square-root factor.

The theorem is conditional. It does not show unconditionally that the reduction cannot exist. Rather, a correct reduction would itself establish a major and unexpected collapse of complexity classes.

2 Preliminaries

2.1 Promise hardness

We use deterministic polynomial-time promise-preserving many-one reductions. If A is a language and $\Pi = (\Pi_Y, \Pi_N)$ is a promise problem, a reduction R satisfies

$$x \in A \Rightarrow R(x) \in \Pi_Y, \quad x \notin A \Rightarrow R(x) \in \Pi_N.$$

We call Π NP-hard if such a reduction exists from an NP-complete language.

Let n denote source length, $M(n)$ the bit length of the emitted lattice instance, and $D(n)$ its rank. Polynomial time implies

$$M(n) \leq n^{O(1)}, \quad D(n) \leq M(n). \quad (3)$$

We do not assume a polynomial lower bound on $D(n)$ unless stated.

2.2 Quasipolynomial and exponential resources

Write

$$\text{qpoly}(n) = 2^{(\log n)^{O(1)}}.$$

The class AM_{qpoly} consists of languages with constant-round public-coin protocols whose verifier time, randomness, and total communication are quasipolynomial. Define PH_{qpoly} by allowing a fixed number of alternations with quasipolynomial resources.

The exponential hierarchy is

$$\text{EH} = \bigcup_{k \geq 1} \Sigma_k^{\text{EXP}}.$$

We use the following published collapse theorem.

Theorem 2.1 (Pavan–Selman–Sengupta–Vinodchandran [PSSV07]) *If $\text{coNP} \subseteq \text{AM}_{\text{qpoly}}$, then*

$$\text{PH}_{\text{qpoly}} \subseteq \text{AM}_{\text{qpoly}}$$

and, by padding,

$$\text{EH} \subseteq \text{AM}_{\text{EXP}}.$$

The standard containment $\text{AM}_{\text{EXP}} \subseteq \Sigma_2^{\text{EXP}} \cap \Pi_2^{\text{EXP}}$, together with $\Sigma_2^{\text{EXP}} \subseteq \text{EH}$, therefore gives

$$\text{EH} = \text{AM}_{\text{EXP}} \subseteq \Pi_2^{\text{EXP}}.$$

Consequently the exponential hierarchy collapses to its second level.

This strengthens an earlier theorem of Selman and Sengupta that obtained only a third-level collapse. The distinction is important: the later argument extends the Boppana–Håstad–Zachos proof to quasipolynomial resource bounds before applying padding.

3 A constant-sensitive coNP barrier at the unit factor

This section sharpens the constants in the verifier of Aharonov and Regev [AR05]. Their published Theorem 1.1 says that $\text{GapCVP}_{c\sqrt{D}}$ lies in $\text{NP} \cap \text{coNP}$ for some absolute $c > 0$; the theorem statement leaves c symbolic, while the proof explicitly establishes the safe choice $c = 100$. The smaller constant below is a new analysis of their verifier, not a value claimed in their paper.

Their normalized periodic Gaussian has a Fourier distribution $\hat{f}$ on the dual lattice with two estimates:

$$\mathbb{E}_{w \sim \hat{f}} \langle u, w \rangle^2 \leq \frac{1}{2\pi} \quad \text{for every unit } u, \quad (\text{A1})$$

$$f(t) \leq 2^{-\Omega(D)} \quad \text{if } \text{dist}(t, L) \geq a\sqrt{D}, \quad a > \frac{1}{\sqrt{2\pi}}. \quad (\text{A2})$$

They also prove an exponentially small tail above $C\sqrt{D}$ for every fixed $C > 1/\sqrt{2\pi}$. These are Lemmas 2.5, 2.6, and 3.1 of [AR05].

Lemma 3.1 (Quadratic support for cosine) For $0 \leq A \leq \pi/2$ and every $y \in \mathbb{R}$,

$$\cos(Ay) \geq \cos A - \frac{A \sin A}{2}(y^2 - 1). \quad (\text{A3})$$

Consequently, if $N^{-1} \sum_i y_i^2 \leq 1$, then $N^{-1} \sum_i \cos(Ay_i) \geq \cos A$.

Proof. The claim is immediate for $A = 0$, and the difference between the two sides is even in y . For $A > 0$ and $y \geq 0$, let

$$g(y) = \cos(Ay) - \cos A + \frac{A \sin A}{2}(y^2 - 1).$$

Then $g(1) = 0$ and $g'(y) = A(y \sin A - \sin(Ay))$. On $0 \leq y \leq 1$, concavity of sine gives $\sin(Ay) \geq y \sin A$, so g decreases to zero. On $1 \leq y \leq \pi/A$, monotonicity of $\sin x/x$ on $(0, \pi)$ gives $\sin(Ay) \leq y \sin A$. Finally, when $y \geq \pi/A$, concavity of sine on $[0, \pi/2]$ gives $(\pi/A) \sin A \geq 2$, while $\sin(Ay) \leq 1$; hence again $g'(y) \geq 0$. Thus $g(y) \geq 0$ everywhere. Averaging (A3) proves the final statement. $\square$

Lemma 3.2 (Constant-sensitive empirical covariance) For every fixed rational $\sigma^2 > 1/(2\pi)$, there is a polynomial $N = D^{O_\sigma(1)}$ and a dual-lattice witness $W = (w_1, \dots, w_N)$ for which

$$\frac{1}{N} WW^\top \preceq \sigma^2 I, \quad (\text{A4})$$

and, for every fixed target satisfying (A2), $N^{-1} \sum_i \cos(2\pi \langle t, w_i \rangle)$ is smaller than any prescribed positive constant threshold.

Proof. Put $\eta = \sigma^2 - 1/(2\pi) > 0$. Sample independently from $\hat{f}$ and replace a sample by zero when its norm exceeds $C\sqrt{D}$. For a fixed unit vector u , the resulting squared projection is in $[0, C^2 D]$ and has expectation at most $1/(2\pi)$ by (A1). Hoeffding makes its empirical mean at most $1/(2\pi) + \eta/2$ except with probability $\exp(-\Omega_\eta(N/D^2))$. Apply this estimate to a fixed sufficiently fine net of the unit sphere. The net has size $\exp(O_\eta(D))$, and the standard positive-semidefinite net inequality

$$\|S\|_{\text{op}} \leq (1 - 2\delta)^{-1} \max_{u \text{ in the } \delta\text{-net}} u^\top S u$$

turns the net bounds into (A4) when $\delta = \Theta(\eta)$. Taking N to be a sufficiently large polynomial of degree greater than three makes the union bound succeed. The Gaussian tail bound shows simultaneously that no sampled vector was actually truncated.

For the fixed target, the cosine average has expectation $f(t)$, which is exponentially small by (A2), and its summands lie in $[-1, 1]$. A second Hoeffding bound puts the average below any fixed positive threshold. The intersection of these events has positive probability, proving the existence of one polynomial-size witness. All retained vectors have polynomially bounded norm and therefore polynomial encoding length, as in the precision analysis of [AR05]. Because σ^2 and the witness entries are rational, the covariance test is an exact polynomial-time rational positive-semidefiniteness check. $\square$

Theorem 3.3 (Constant-sensitive Aharonov–Regev sharpening) For every fixed $\varepsilon > 0$,

$$\text{GapCVP}_{(2/\pi+\varepsilon)\sqrt{D}} \in \text{NP} \cap \text{coNP}. \quad (\text{A5})$$

Consequently $\text{GapCVP}_{\sqrt{D}} \in \text{NP} \cap \text{coNP}$.

Proof. NP membership is standard, so it remains to certify far instances. Fix a rational $\sigma^2 > 1/(2\pi)$ and require (A4). If the target is within distance b of the lattice, periodicity reduces it to a vector x with $\|x\| \leq b$. The case $x = 0$ is immediate. Otherwise put

$$y_i = \frac{\langle x, w_i \rangle}{\sigma \|x\|}, \quad A = 2\pi\sigma\|x\|.$$

Equation (A4) gives $N^{-1} \sum_i y_i^2 \leq 1$. If $b < 1/(4\sigma)$, then $A < \pi/2$, and Lemma 3.1 gives

$$\frac{1}{N} \sum_i \cos(2\pi \langle x, w_i \rangle) \geq \cos(2\pi\sigma\|x\|) \geq \cos(2\pi\sigma b) > 0. \quad (\text{A6})$$

Choose a rational verification threshold strictly between zero and the last quantity. No witness passing (A4) can cross this threshold on a close instance. Conversely, the empirical-covariance lemma and (A2) provide a witness below the threshold whenever the distance is at least $a\sqrt{D}$ with $a > 1/\sqrt{2\pi}$.

After rescaling the close radius b to one, the approximation factor is $(a/b)\sqrt{D}$. Sending

$$a \downarrow \frac{1}{\sqrt{2\pi}}, \quad \sigma^2 \downarrow \frac{1}{2\pi}, \quad b \uparrow \frac{\sqrt{2\pi}}{4}$$

makes $a/b \downarrow 2/\pi$. Fixed parameters can therefore be chosen with $a/b < 2/\pi + \varepsilon$, and promise monotonicity gives (A5). Choosing $\varepsilon < 1 - 2/\pi$ yields the unit-factor statement. $\square$

Corollary 3.4 (Exact square-root hardness consequence) *If $\text{GapCVP}_{\sqrt{D}}$ is NP-hard under deterministic promise-preserving many-one reductions, then $\text{NP} = \text{coNP}$.*

Proof. Theorem 3.3 puts the target problem in coNP. Composing a Karp reduction from SAT with its verifier gives $\text{NP} \subseteq \text{coNP}$; taking complements gives the reverse containment. $\square$

Thus the exact unit-factor hardness statement is false under $\text{NP} \neq \text{coNP}$. Without a separation assumption it remains logically open: the containment proves the collapse consequence, not the unconditional nonexistence of a reduction. Independently, the universal coAM protocol used below gives the EH-noncollapse and AMETH barriers without any leading-constant ambiguity.

4 The quantitative coAM protocol

Aggarwal et al. prove the following tradeoff.

Theorem 4.1 (Aggarwal et al.) *Corollary 3.3 of [ABB⁺26] states that for every $\gamma = \gamma(D) \geq 1$, rank- D Euclidean GapCVP_γ has a constant-round coAM protocol of complexity*

$$2^{O(D/\gamma^2)} \text{poly}(M), \quad (4)$$

where M is the input encoding length.

The underlying private-coin protocol performs

$$N_2 = 10D^{3/2}(1 - 1/\gamma^2)^{-(D+1)/2} \quad (5)$$

independent trials. In each trial Arthur chooses a hidden uniform bit and a uniform point in one of two translated Euclidean balls; Merlin must identify the hidden translate. Generic private-to-public-coin and round-reduction transformations incur polynomial overhead, preserving the exponent in (4).

For $\gamma(D) = \sqrt{D}/f(D)$, (4) becomes

$$2^{O(f(D)^2)} \text{poly}(M). \quad (6)$$

Equation (6) is the only lattice-specific input needed below.

5 The hierarchy-collapse barrier

We first state the source-sensitive theorem.

Theorem 5.1 (General reduction barrier) *Suppose there is a deterministic polynomial-time promise-preserving many-one reduction from SAT instances of length n to Euclidean GapCVP_γ instances of rank $D(n)$ and encoding length $M(n)$. If, for some fixed a ,*

$$\frac{D(n)}{\gamma(D(n))^2} \leq (\log n)^a \quad (7)$$

for all sufficiently large n , then

$$\text{EH} = \text{AM}_{\text{EXP}} \subseteq \Pi_2^{\text{EXP}}.$$

In particular, the exponential hierarchy collapses to its second level.

Proof. By Theorem 4.1, the target promise problem has a coAM protocol of complexity

$$2^{O(D(n)/\gamma(D(n))^2)} \text{poly}(M(n)).$$

Equations (3) and (7) make this quantity quasipolynomial in n . Composing the verifier with the deterministic reduction gives

$$\text{SAT} \in \text{coAM}_{\text{qpoly}},$$

and complementing gives $\text{TAUT} \in \text{AM}_{\text{qpoly}}$. Since TAUT is coNP -complete under polynomial-time many-one reductions,

$$\text{coNP} \subseteq \text{AM}_{\text{qpoly}}.$$

The conclusion follows from Theorem 2.1. □

Corollary 5.2 (Fixed-polylogarithmic loss) *For every fixed $C > 0$, if*

$$\text{GapCVP}_{\sqrt{D}/\log^C D}$$

is NP-hard under polynomial-time promise-preserving many-one reductions, then the exponential hierarchy collapses to its second level.

Proof. Here

$$\frac{D}{\gamma(D)^2} = \log^{2C} D.$$

Since $D(n) \leq M(n) \leq n^{O(1)}$, this is $\log^{O(1)} n$, so Theorem 5.1 applies. □

Corollary 5.3 (Conditional impossibility) *Assume that the exponential hierarchy does not collapse to its second level. Then, for every fixed $C > 0$, Euclidean $\text{GapCVP}_{\sqrt{D}/\log^C D}$ is not NP-hard under deterministic polynomial-time promise-preserving many-one reductions.*

This conclusion is independent of the internal form of the reduction. It applies equally to reductions built from codes, PCPs, Fourier statistics, group actions, geometric gadgets, or any other polynomial-time compiler.

6 The exact asymptotic frontier of this argument

Write the approximation factor as

$$\gamma(D) = \frac{\sqrt{D}}{f(D)}, \quad (8)$$

where $f(D)$ is the loss from the square-root scale. Theorem 5.1 becomes the following exact criterion.

Proposition 6.1 (Protocol-frontier criterion) *A polynomial-time reduction with loss f implies the second-level exponential-hierarchy collapse whenever*

$$f(D(n))^2 = \log^{O(1)} n. \quad (9)$$

In particular, every dimension-only bound

$$f(D) = \log^{O(1)} D \quad (10)$$

satisfies (9) for every polynomial-time reduction.

The converse is deliberately not claimed. If $f(D)$ is superpolylogarithmic in D , condition (9) may still hold when the reduction emits unusually small rank relative to source length. To state a clean dimension-only boundary, one may impose the natural rank-regularity condition

$$D(n) \geq n^\delta \quad (11)$$

for some fixed $\delta > 0$.

Corollary 6.2 (Rank-regular boundary) *Under (11), the protocol in (4) has quasipolynomial complexity precisely when*

$$f(D) = \log^{O(1)} D$$

along the output ranks, up to the usual eventual-bound interpretation. Thus the first dimension-only region not covered by the EH-noncollapse argument consists of losses not bounded by any fixed polylogarithm. The near-square-root portion of this region is

$$\log^C D = o(f(D)) \text{ for every fixed } C, \quad f(D) = D^{o(1)}, \quad \gamma(D) = \frac{\sqrt{D}}{\log^{\omega(1)} D}. \quad (12)$$

There is no smallest superpolylogarithmic function. Equation (12) describes a boundary class, not one canonical approximation factor. Examples include

$$f(D) = \exp((\log \log D)^{1+\varepsilon}) \quad \text{and} \quad f(D) = \exp(\sqrt{\log D}).$$

Both give $\gamma(D) = D^{1/2-o(1)}$, which is asymptotically larger than D^c for every fixed $c < 1/2$, while the protocol in (6) is superquasipolynomial.

Failure of the present argument in (12) is not evidence that hardness exists there. It says only that the published quantitative coAM protocol no longer places the composed source problem in AM_{qpoly} .

7 The stronger AMETH boundary

Aggarwal et al. define AMETH as the assertion that, for some $\varepsilon > 0$, 3-TAUT has no constant-round public-coin protocol of complexity $2^{\varepsilon n}$. The same composition gives a stronger but less structural conditional barrier. Here, unlike in Sections 2–5, n denotes the number of variables. We restrict the proposed reduction to 3-CNF formulas with $O(n)$ clauses and let $D(n)$ be the largest emitted rank on those inputs. Their encoding length is $O(n)$; the standard sparsification lemma [IPZ01] lifts any $2^{o(n)}$ protocol for this sparse restriction to a $2^{\delta n + o(n)}$ protocol for general 3-TAUT, for every fixed $\delta > 0$.

Proposition 7.1 (AMETH barrier) *Under AMETH, no reduction as in Theorem 5.1 can satisfy*

$$\frac{D(n)}{\gamma(D(n))^2} = o(n). \quad (13)$$

on all 3-CNF inputs with n variables and $O(n)$ clauses. Equivalently, for (8), it cannot satisfy

$$f(D(n))^2 = o(n). \quad (14)$$

Proof. Equations (4) and (13) give a $2^{o(n)}$ constant-round protocol for sparse 3-TAUT. Apply the sparsification lemma and run the resulting sparse protocols in parallel. For every fixed $\delta > 0$, this gives a constant-round protocol of complexity $2^{\delta n + o(n)}$ for general 3-TAUT. Taking δ below the constant in AMETH yields a contradiction. $\square$

If $D(n) \leq n^K$ for fixed K and $f(D) = D^{o(1)}$, then $f(D(n))^2 = n^{o(1)} = o(n)$. Hence AMETH excludes the entire $D^{1/2-o(1)}$ approximation regime for polynomial-output reductions, not only fixed-polylogarithmic losses. The superpolylogarithmic strip (12) is therefore open only relative to the weaker EH-noncollapse assumption; it is already forbidden by AMETH.

For a fixed-power factor $\gamma(D) = D^c$, the protocol exponent is D^{1-2c} . If $D(n) \leq n^K$, AMETH forces the efficiency ceiling

$$K(1 - 2c) \not\leq 1, \quad \text{or heuristically} \quad c \leq \frac{1}{2} - \frac{1}{2K}, \quad (15)$$

unless constants or lower-order behavior keep (13) from holding. Thus hardness for every fixed $c < 1/2$ can coexist with the barrier when approaching $1/2$ requires an unbounded output-degree parameter; a uniform $D^{1/2-o(1)}$ theorem cannot.

8 Why the protocol does not give a deterministic algorithm

The conclusion of Corollary 5.2 is an interactive-proof collapse, not a quasipolynomial-time algorithm for GapCVP. This distinction cannot be removed by naively enumerating Arthur’s random coins.

The private protocol underlying (4) uses N_2 independent hidden bits even before accounting for the random convex-body samples. Therefore its written implementation consumes at least N_2 random bits. At the near-square-root endpoint, substituting $\gamma = \sqrt{D}/f(D)$ into (5) gives

$$\log N_2 = \log 10 + \frac{3}{2} \log D + \frac{1}{2} f(D)^2 + O\left(\frac{f(D)^2 + f(D)^4}{D}\right). \quad (16)$$

Thus $N_2 = 2^{\log^{O(1)} D}$ for polylogarithmic f , and exhaustive enumeration costs

$$2^{\Omega(N_2)},$$

not quasipolynomial time. Corollary 3.3 of [ABB⁺26] invokes generic Goldwasser–Sipser [GS89] and Babai–Moran [BM88] transformations with polynomial overhead; it does not replace the experiment by a pairwise-independent test with a logarithmic seed.

More precisely, let ℓ be the amplified private random-tape length in the Goldwasser–Sipser simulation. Its explicit public-coin implementation uses full random linear maps and at least $2\ell^2$ public random bits at an intermediate stage. Since $\ell \geq N_2$, enumerating those public strings already costs $2^{\Omega(N_2^2)}$. The final Babai–Moran transformation is stated only with polynomial resource overhead and supplies no alternative polylogarithmic seed bound. Pairwise-universal hashing in the proof therefore does not amount to a short-seed derandomization.

Nor can a standard pseudorandom generator simply be substituted. In a public-coin protocol Merlin’s response depends on the revealed coins, and soundness quantifies over all prover strategies. Fooling a fixed space-bounded computation is not by itself a derandomization of this interactive maximum. Accordingly, this note claims $\text{coNP} \subseteq \text{AM}_{\text{qpoly}}$ under the hypothetical hardness reduction, but not $\text{coNP} \subseteq \text{QP}$ and not a deterministic quasipolynomial-time algorithm for GapCVP .

One may obtain the standard nonuniform containment

$$\text{coAM}_{\text{qpoly}} \subseteq \text{coNP}/\text{qpoly}$$

by fixing a quasipolynomial-size advice list of public strings after amplification and a union bound over all inputs of a given length. The advice slash is essential; this is not a uniform seed-enumeration algorithm and is not used in our main theorem.

9 Relation to fixed-exponent hardness

Mira (@Mira__Mira_) proves NP-hardness at D^c for every fixed $c < 1/2$ [Mir26]. This theorem does not diagonalize to (1): each fixed exponent corresponds to polynomial loss

$$f(D) = D^{1/2-c},$$

whereas (1) has only polylogarithmic loss. A family of separate reductions, one for each fixed c , need not yield a single polynomial-time reduction with $c = c(D) \rightarrow 1/2$. Typically the polynomial degree of the output size depends on c and becomes unbounded as $c \rightarrow 1/2$.

Under EH noncollapse, Corollary 5.2 says that this distinction is not merely a defect of one known construction: no polynomial-time Karp reduction can reach the fixed-polylogarithmic-loss endpoint. It does not rule out an improvement into the superpolylogarithmic strip (12).

At exponent $1/2$, the leading constant matters. Theorem 3.3 gives a coNP barrier for every fixed factor $\lambda\sqrt{D}$ with $\lambda > 2/\pi$, including the exact unit factor. For smaller positive constants, the polynomial-time coAM barrier still gives the usual polynomial-hierarchy collapse consequence, but the present constant-sensitive argument does not claim coNP membership.

10 Conclusion

The quantitative coAM protocol for Euclidean GapCVP and the resource-scaled interactive-proof collapse theorem together give a universal conditional boundary:

$$\text{NP-hardness at } \sqrt{D}/\log^{O(1)} D \implies \text{the exponential hierarchy collapses to level two.}$$

This makes any such hardness proof simultaneously a hierarchy-collapse proof. Assuming EH noncollapse, the fixed-polylogarithmic-loss endpoint is closed.

At the exact factor $\sqrt{D}$, the conclusion is stronger:

$$\boxed{\text{NP-hardness of GapCVP}_{\sqrt{D}} \implies \text{NP} = \text{coNP}.}$$

This uses the verified constant-sensitive sharpening in Theorem 3.3; the published Aharonov–Regev theorem itself states only the existence of an absolute constant and displays the safe value 100.

The next unresolved asymptotic target, for rank-regular reductions and under that assumption alone, is hardness with a loss that is simultaneously superpolylogarithmic and subpolynomial, schematically

$$\frac{\sqrt{D}}{\log^{\omega(1)} D}.$$

This strip lies strictly beyond every fixed power D^c , $c < 1/2$, but outside the quasipolynomial-coAM zone. Establishing hardness there would require a new reduction; excluding it would require an assumption or upper bound stronger than EH noncollapse.

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